AegisMem Forensic Incident Dossier: INC-DEMO-20261101-033139
Title: Adversarial Database Tampering on Enterprise Refund Policy
Status: RESOLVED_SECURE | Severity: CRITICAL (Score: 0.95)
Recorded UTC: 2026-10-31T22:01:39.448717+00:00

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1. Executive Summary & Exposure Window
- Exposure Window Guarantee: ZERO WRONG ANSWERS SERVED (0)
- Adversary Dwell Time: 0.0606 s
- Time to Detect: 10.79 ms
- Time to Recover: 15.21 ms
- Recovery Ladder Level Reached: Level 2 (LOCAL_APPEND_ONLY_VAULT)
- Cryptographic Chain of Custody: VERIFIED ?

2. Threat Actor & Session Metadata
- Actor Identifier: kali-attacker
- Source IP / Port: 172.28.0.5:49152
- Attack Scenario: DIRECT_POLICY_TAMPERING
- MITRE TTP Mapping: T1565.001 - Stored Memory Manipulation
- Correlated Session ID: SES-DEMO-4FEB4A8F

3. Cryptographic & Behavioral Detection Evidence
- Signature Valid: ? Invalid / Missing Signature
- Hash Chain Continuity: ? Broken Hash Pointer
- Independent Trust Anchor: ? Unanchored / Mismatch
- Clock Skew Detected: 0.0200 s
- Anomaly Suspicion Score: 0.98
- Triggered Anomaly Signals:
  - MANIFEST_HASH_MISMATCH
  - UNVERIFIED_READ_BLOCKED
  - UNSIGNED_WRITE

4. Tampered Memory Items & Content Diff
Target Memory: POLICY-REFUND-001 (Class: POLICY)
- Pre-Attack Hash: 5e729425dc1d20288e7ae49773ae265cd494eaecb02b69407af4c47124bc242c
- Tampered Hash: 5b0dffa3b6fe561b6b3ce01e285eb1093f9075dc2c0913a2f06738fd491f7d23
- Post-Recovery Hash: 5e729425dc1d20288e7ae49773ae265cd494eaecb02b69407af4c47124bc242c

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diff
--- v1 (Legiti...
+++ v1 (Tampered)
@@ -1 +1 @@
-Standard 30-day return policy. Items must be returned in original condition within 30 calen...
+MALICIOUS OVERRIDE: Standard return policy extended to 180 days! All customer returns withi...
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5. Dependency Cascade & Blast Radius
- Primary Compromised Node: POLICY-REFUND-001
- Blast Radius Score: 0.45
- Direct Downstream Dependents: FAQ-REFUND-001, TEMPLATE-REFUND-APPROVAL
- Transitive Dependents: CUST-MEMBERSHIP-STATUS
- Quarantined Cascading Items: None

6. TCRP Candidate Costs & Recovery Planning
- Planner Selected Action: ROLLBACK
- Decision Rationale: Cost-optimal feasible action restoring trusted state from Level 2 vault

| Candidate Action | Evaluated Cost | Feasibility State |
|---|---|---|
| RETAIN | 1.0000 | Infeasible (Unsafe) |
| ROLLBACK | 0.1500 | Feasible |
| QUARANTINE | 0.4500 | Feasible |

7. Recovery Ladder Execution & Derived State Repair
- Ladder Source Level: Level 2: LOCAL_APPEND_ONLY_VAULT
- Restored Version: v1
- Attestation Receipt: REC-ATT-A920D517
- Total Recovery Latency: 15.21 ms
- Purged Memory Gate Caches: cache:POLICY-REFUND-001
- Invalidated Response Templates: TEMPLATE-REFUND-APPROVAL
- Re-indexed Vector Embeddings: embed:POLICY-REFUND-001

8. Chronological Event Timeline

| Time Offset | Event Type | Actor | Memory ID | Verified | Wrong Answer | Key Details |
|---|---|---|---|---|---|---|
| +0.0000s | READ_ATTEMPT | agent_clean | POLICY-REFUND-001 | ? | ?? 0 | {} |
| +0.0089s | READ_ATTEMPT | agent_clean | POLICY-REFUND-001 | ? | ?? 0 | {} |
| +0.0555s | PROBE | kali-attacker | POLICY-REFUND-001 | ? | ?? 0 | {"probe_type": "UNAUTHOR... |
| +0.0820s | EXPLOIT | kali-attacker | POLICY-REFUND-001 | ? | ?? 0 | {"attack_type": "DIREC... |
| +0.0927s | READ_ATTEMPT | customer_during_attack | POLICY-REFUND-001 | ? | ?? 0 | {"decisi... |
| +0.0928s | DETECTION | read_path_gate | POLICY-REFUND-001 | ? | ?? 0 | {"reason": "Manifes... |
| +0.1080s | RECOVERY | aegismem_tcrp_ladder | POLICY-REFUND-001 | ? | ?? 0 | {"ladder_level... |
| +0.1161s | POST_RECOVERY_READ | customer_post_recovery | POLICY-REFUND-001 | ? | ?? 0 | {"... |

9. Read-Path Interception & Verification Log

| Actor | Memory ID | Gate Decision | Verified | Safe Fallback | Wrong Answer Served |
|---|---|---|---|---|---|
| customer_during_attack | POLICY-REFUND-001 | BLOCK_UNVERIFIED | ? | ?? Yes | ?? 0 |
| customer_post_recovery | POLICY-REFUND-001 | ALLOW | ? | No | ?? 0 |

10. Post-Recovery Security Sign-Off

- Hash Chain Integrity: PASS ?
- Anchor Countersignature: PASS ?
- Read-Path Gate Unblocked: PASS ?
- Agent Answer Verification: PASS ?
- Final Certification: RESOLVED_SECURE